(GSiC)

Government Services Integration Cluster

Ontario Business Registry (OBR)

Ontario

Ontario Business Registry API – Technical Assessment Guide for Transactional Services Licensees

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Acknowledgments

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1 Introduction

1.1 Purpose

This document will provide a technical overview of the APIs offered by the Ontario Business Registry (OBR) system to aid potential licensees in completing a high-level assessment of the activities required to consume these APIs successfully. This document does not delve into the specific, low-level technical details of the APIs and related security controls. Such information will be made available to licensees once the necessary licensing agreements have been fully executed between the Ministry and the licensees.

Note that a licensee can be any private-sector or public-sector organization – whether for-profit or not-for-profit – that intends to submit business transactions into the OBR system on behalf of other parties (such as their own clients).

1.2 Audience

This document is intended for technical architects, systems analysts and/or software developers with licensee organizations that wish to integrate with the secure OBR API interface that The Ministry has made available to external partners. Readers should review and be knowledgeable of the following:

- Ontario Business Registry (https://www.ontario.ca/page/ontario-business-registry)
- JSON standards, used in REST transactions (https://www.json.org/json-en.html)
- REST specifications using OpenAPI 3 standard (https://swagger.io/specification/v3/)
- Public key certificates using X.509 format (https://www.itu.int/rec/T-REC-X.509-201910-I/en)

1.3 Context

The Ontario Business Registry is an online registry system that makes it easy for businesses and not-for-profit corporations to interact with government through a wide range of business-related transactions such as registering a new business and dissolving an existing one. The registry system offers a collection of secure APIs to external partners to enable fast and efficient retrieval and submission of information for businesses that currently operate (or formerly operated) in Ontario. Prospective licensees may choose to consume these APIs as part of the offerings that they deliver to their clients.

While the list of business transactions that can be executed via OBR APIs is extensive, not every OBR transaction can be performed through an API call. Therefore, the APIs are intended to be

used in conjunction with The Ministry's external-facing partner website known as the "Partner Portal". All OBR business services can be executed via the Partner Portal. All licensees who opt to consume the OBR APIs must also be configured for Partner Portal access.

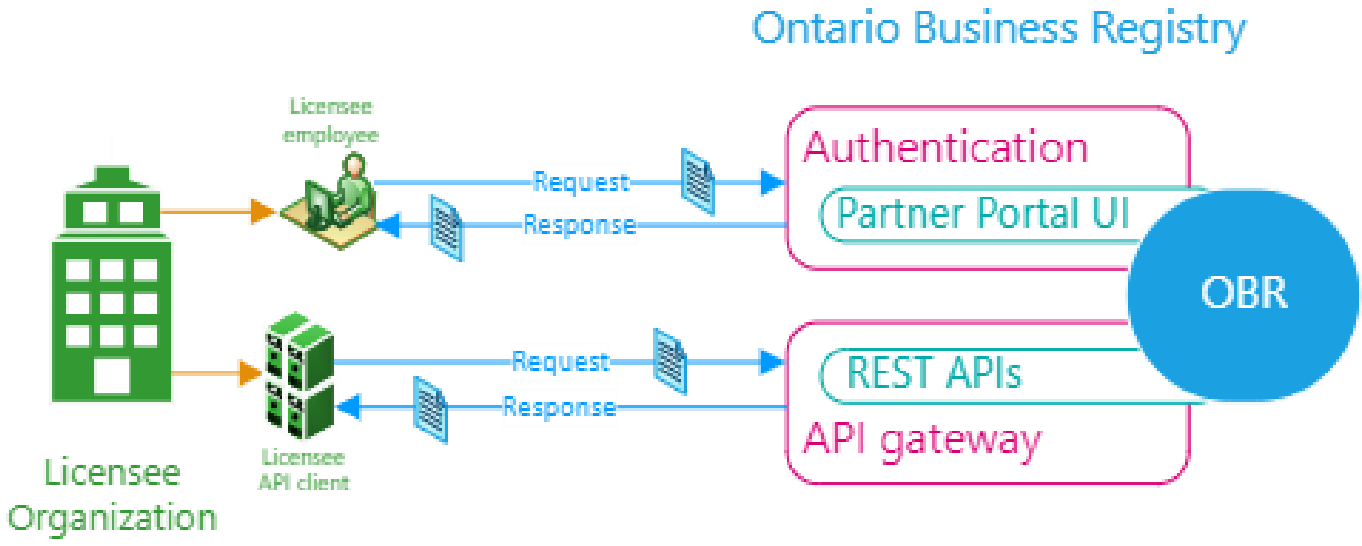


Figure 1: Ontario Business Registry high level view

The OBR APIs share the following characteristics:

- All OBR APIs use the REST protocol (JSON messages) over HTTPS to search, update and lodge (register) information for a new or existing business in Ontario.
- The APIs are secured through X.509 certificates. Only licensees who have worked with the Ministry to set up a valid security configuration will be able to access the APIs successfully.
- The APIs are intended for system-to-system communications.

This document will assist prospective licensees in identifying and estimating the technical efforts required for them to consume the offered APIs. The Ministry's responsibilities are limited to:

- making the suite of secure APIs available to licensees in both a non-production environment for testing purposes (UAT environment) and a live production environment;
- assisting partners with configuring their security and connectivity; and
- confirming they can call the APIs successfully.
- Providing updates schemas and specifications

Once a licensee has confirmed that they can call the APIs successfully, all other technical and business tasks related to delivery of services to the licensee's own clients (including creation of any value-add products or services) are the responsibility of the licensee.

1.4 Glossary

Term	Description
API	Application Programming Interface
APIC	IBM API Connect
CA	Certificate Authority
GSIC	Government Services Integration Cluster, the internal IT services organization that supports the OBR within the Ministry
HTTP	Hypertext Transfer Protocol
ITU	International Telecommunications Union
JSON	JavaScript Object Notation
Keystore	A password protected file containing cryptographic key and certificates (such as the X.509), used to establish a secure connection. These files come with extensions such as .jks, .p12 or .pfx
MPBSDP	Ministry of Public and Business Service Delivery and Procurement
NAICS	North American Industry Classification System, a classification of business establishments by type of economic activity (process of production) used by government and business in Canada, Mexico, and the USA.
NUANS	Newly Upgraded Automated Name Search, an automated search service offered by *Innovation, Science and Economic Development Canada* for comparison of proposed corporate names or trademarks
OBR	Ontario Business Registry
ONBIS	Ontario Business Information System. This is the legacy name for the OBR application. "ONBIS" has been deprecated, and "OBR" should be used instead.
OPS	Ontario Public Service
ReadyAPI Postman	API testing tools for applications that use REST-based message formats.
REST	Representational State Transfer
TLS	Transport Layer Security
UI	User Interface
X.509	ITU standard that defines the format of public key certificates.

2 APIs

2.1 Overview

An API is a service offered by an electronic device to one or more other electronic devices via the Internet. The devices intercommunicate using standard protocols and data formats such as HTTP, XML, JSON, etc. The Ontario Business Registry system offers a suite of secure REST APIs to external partners to enable fast and efficient retrieval and submission of information for businesses that currently operate (or formerly operated) in Ontario.

2.2 Services offered

The table below provides a high-level description of the APIs available in the OBR. When reviewing the API list, please also note the following:

- The term "entity" is used to refer to *any* kind of business entity in the registry, such as a corporation, a partnership, a sole proprietorship, a not-for-profit, etc.
- Some business transactions require calls to more than one API to complete a single transaction.
- Some APIs require payment of a statutory fee to execute the transaction successfully. For more information on payment, please refer to Section 2.4.3.
- Some APIs require that the caller have "authority" over a business entity to execute a transaction for that entity. If the caller attempts to execute one of these APIs for an entity over which it does *not* have authority, either the transaction will fail or it will reply with limited information about the entity. For more information on authority, please refer to the Partner Portal Onboarding Guide for Intermediary Users.

Service Name	Description
General APIs	
Check Entity Name	This API allows licensees to submit a request to check an entity name against OBR-specific validation rules for names and to determine for certain corporation types if a legal opinion is required. Refer to the following link for information on Notice of filing requirements: https://www.ontario.ca/page/ontario-business-registry-all-services#section-6.

	A NUANS validation is not included in this service.
Entity Search	This API allows licensees to search for registered entities. If more than 50 results are found, only the first 50 results will be returned along with an informational message.
Get Authority	This API allows licensees to gain authority for an entity using the Company Key.
Get Entity	This API allows licensees to retrieve the current details of a single entity.
Get Entity documents list	This API allows licensees to retrieve the document list for a single entity. It is a prerequisite for the "Request Document Copies" service. After a registration or maintenance business service is performed, any subsequent documents take a few minutes to generate. Please allow some time for the generation of new PDF documents.
Retrieve Entity Profile Report	This API allows licensees to retrieve the Profile Report details (latest version) of a single entity, in the response message payload. "Request Entity Profile Report" must be performed prior to submitting this request.
Retrieve Service Documents	This API allows licensees to retrieve the service-specific documents. Some of the Services which must be called prior to calling this service include: • Request Document Copies • Request Certificate of No Match • Request Certificate of Status • Request Entity Profile Report
Request Document Copies	This API allows licensees to request document copies of a single entity. The Get Entity documents list must be performed to determine the documents available for an entity. If the document is not already available in OBR system, a request will be sent to back-office staff to retrieve the documents from the archives.
Request Certificate of No Match	This API allows licensees to request a certificate of no match for a single entity.

Request Certificate of Status	This API allows licensees to request a certificate of status for a single entity.
Request Entity Profile Report	This API allows licensees to request a profile report for a single entity.
Business Corporation APIs	
Annual Return for an Ontario Corporation	This API allows authorized licensees to submit an Annual Return for an Ontario corporation. If there is information to be updated, then only the new values need to be submitted. Otherwise, only the Entity Identifier, Year of Annual Return, and Change Certification need to be submitted. Prior to using this API, you must perform "Get Entity" to retrieve the latest corporation information.
Incorporate an Ontario Business Corporation	This API allows licensees to submit incorporation application for an Ontario Business Corporation entity type.
Initial Return for an Ontario Corporation	This API allows authorized licensees to submit an Initial Return for an Ontario corporation. If there is information to be updated, then only the new values need to be submitted. Otherwise, only the Entity Identifier and Change Certification need to be submitted. Prior to using this API, you must perform "Get Entity" to retrieve the latest corporation information.
Notice of Change for an Ontario Corporation	This API allows authorized licensees to submit a Notice of Change for an Ontario corporation.
Register a Business Name for a Corporation	This API allows authorized licensees to register a business name for a corporation.
Renew a Business Name for a Corporation	This API allows authorized service providers to renew a business name for a corporation.
Sole Proprietor APIs	
Register a Business Name for a Sole Proprietorship	This API allows authorized licensees to register a business name for a sole proprietorship.

Renew a Business Name for a Sole Proprietorship	This API allows authorized licensees to renew a business name for a sole proprietorship.
General Partnership APIs	
Register a Firm Name for a General Partnership	This API allows authorized licensees to register a firm name for a general partnership.
Renew a Firm Name for a General Partnership	This API allows authorized licensees to renew a firm name for a general partnership.

2.3 Prerequisites for consumption of APIs

The following prerequisite tasks must be completed before licensees can call any OBR APIs successfully:

- The licensee must be defined as a known Organization within the Business Registry system. Refer to the Partner Portal Onboarding Guide for Intermediary Users for more details regarding the Organization setup process.
- The licensee organization must use X.509 certificates issued by the Government of Ontario.
    - Note that licensees will be required to use different certificates for production vs. non-production environments.
    - A certificate used in a testing environment (e.g., UAT environment) will not be permitted in a production environment.
    - See section 5.2 of this document for more information on how OBR uses certificates to secure the APIs.
- The licensee organization must use the client ID issued by the Government of Ontario. This client ID must be included in all API transactions.
- The organization must provide to the Ministry the public IP address(es) from which all calls to OBR APIs will originate. These IP addresses will be added to an 'allow access' list and only calls from these addresses will be permitted to pass through the OPS gateway. For more information regarding IP filtering, refer to section 5.3 of this document.

2.4 API architecture

2.4.1 API standards

OBR APIs conform to the OpenAPI 3 REST specification as per the IBM and Swagger definitions. The table below lists key technology standards utilized by OBR.

Category	Standard being used
Transport	HTTP 1.2 HTTP over TLS
Messaging	JSON messages via REST
Description	OpenAPI 3 specification
Security	TLS 1.3
Electronic Signatures	X.509 certificates issued by the Government of Ontario
Authentication	Each API call must contain specific header elements which uniquely identify an organization based on the X.509 Certificate (see section 5.2)
Firewall Access	OBR will allow access to a limited preconfigured range of IP addresses from licensees (see section 5.3)

2.4.2 API usage policies

To ensure the ongoing stability and reliability of the OBR system, all licensees will be subject to the following usage policies.

2.4.2.1 Rate limiting

Rate limiting ensures that only a predefined maximum number of calls will be accepted within a defined time period, e.g., 1000 calls per hour. When the limit is reached, no more calls are accepted from that licensee until the beginning of the next time period. In this example, if a licensee makes 1000 calls in the first 20 minutes, they cannot complete any more calls until the hour has expired.

Rate limits are defined and enforced by the OBR API gateway. For more information about OBR rate limits, refer to section 3.4. The source of truth for the specific rate limits can be found in the latest version of the OBR Web Services Addendum.

2.4.2.2 Burst limiting

Burst limiting is similar to rate limiting but is applied to a very small time period (typically in multiples of seconds or minutes), e.g., 90 calls per minute. When a burst limit is reached, calls are not accepted until the enough time has passed to accommodate more API calls within the burst limit timeframe. As time progresses, the historical count of transactions over time reduces. Further API calls continue to be accepted until the overall rate limit is reached. Burst limiting prevents usage spikes and ensures that the rate limit is evenly spread across its overall time period.

Burst limits are defined and enforced by the OBR API gateway. For more information about OBR rate limits, refer to section 3.4. The source of truth for the specific burst limits can be found in the latest version of the OBR Web Services Addendum.

2.4.2.3 Retry policy

All licensees will be required to adhere to the Ministry's API Retry Policy. The Retry Policy applies in both timeout scenarios (where no response is returned by the application within a predefined time limit) and non-timeout scenarios (where a response is received but its JSON payload is unexpected or otherwise malformed).

Too many retries calling a slow or unresponsive API (or too short an interval between retries) can have an adverse impact on the target service and potentially on the system as a whole. The service may be prevented from recovering from its overloaded state and it may continue to block and/or refuse requests.

The retry policy is designed to impose limits on the number of retries so that load on the system can be managed appropriately. Licensees are limited to a maximum of retries of the *same* transaction with a wait-time between each unsuccessful attempt. After the maximum number of retries, the transaction should be abandoned, and reported to the Ministry so that the issue can be managed through Ministry's formal incident management process.

Any licensee who does not adhere to the Retry Policy and whose usage of the APIs is discovered to be having an adverse impact on the stability or performance of the OBR system is at risk of having their access revoked. Necessary steps when a licensee is discovered not adhering to the Retry Policy include (but are not limited to):

- representatives from both the licensee organization and the Ministry have conducted a post-mortem of the observed behaviour and its impact on the Business Registry;
- the licensee has committed to the Ministry's satisfaction that:

- appropriate steps have been taken to rectify any shortcomings in the licensee's adherence to the Retry Policy, and;
- similar destabilizing usage of the APIs will not be repeated.

2.4.3 Payment

Some OBR transactions require payment of a statutory fee to complete the transactions successfully. However, payment details cannot be submitted as part of the message payload for any OBR API call. Instead, payment of statutory fees is made possible by means of an OBR "Deposit Account".

A Deposit Account can be thought of as similar in behaviour to a pre-paid credit card in that:

- it can be preloaded with funds via deposits; and
- it can pay for statutory fees through payments that are triggered automatically by successful completion of a business transaction that is subject to a fee.

The balance of funds available in the Deposit Account will be drawn down gradually as the licensee submits transactions that incur statutory fees. Therefore, the Account will need to have funds replenished on an ongoing basis as the licensee completes transactions over time. If a licensee attempts to submit a transaction that requires payment and there are insufficient funds available in the Deposit Account, the transaction will fail and the API response will contain a "Payment incomplete" message.

Each licensee organization must have its own Deposit Account that will be preconfigured by the Ministry as part of the initial setup process. The procedure to accept funds and credit the balance of a Deposit Account is managed by the Ministry.

2.4.3.1 API calls that incur a statutory fee

Calls to the following APIs will automatically deduct a preconfigured statutory fee from a licensee's Deposit Account:

General APIs:

- Request Document Copies
- Request Certificate of No Match
- Request Certificate of Status
- Request Entity Profile Report

Business Corporation APIs:

- Incorporate an Ontario Business Corporation
- Register a Business Name for a Corporation
- Renew a Business Name for a Corporation

Sole Proprietor APIs:

- Register a Business Name for a Sole Proprietorship
- Renew a Business Name for a Sole Proprietorship

General Partnership APIs:

- Register a Firm Name for a General Partnership
- Renew a Firm Name for a General Partnership

Refer to https://www.ontario.ca/page/cost-time-required-to-register-change-search-for-business-name-corporation-not-for-profit#section-2 for more details regarding statutory fees for specific transactions/business services.

2.4.4 Date and time values

Some APIs will include a date/time value in the request and/or response. Dates and times are formatted as per the ISO 8601 rules. (Refer to the ISO 8601 website for more information.)

Date/time values will be expressed in Eastern Standard Time/Eastern Daylight Time, as appropriate.

Some examples:

```
Date with time and time zone:   2022-11-18T15:30:00-05:00
Date with time zone:            2022-11-18-05:00
Date only:                      2022-11-18
```

2.5 How OBR APIs work together with the OBR Partner Portal

While the list of business transactions that can be completed via the OBR APIs is extensive, not every OBR transaction can be performed through an API call currently. Therefore, the APIs are intended to be used in conjunction with the OBR Partner Portal:

- *All* OBR business services can be executed via the Partner Portal site.
- The OBR APIs correspond to a subset of the same business services that can be submitted via the Partner Portal.
- All licensees who opt to consume the OBR APIs must also be configured for Partner Portal access.
- Depending on which transaction a licensee is attempting to execute, it is possible that the transaction may require completion via the Partner Portal instead of an API call.
- When requesting entity data via an API call, the data presented in the REST response message will be in JSON format, however it is the same data that is displayed to users via the Partner Portal UI.

3 API gateway

3.1 Overview of API Connect and the APIC Developer Portal

The Ministry utilizes IBM's API Connect (APIC) gateway to group together APIs into collections known as "Products". The Products are then used to manage the versioning, life cycle, access control and consumption of the APIs they contain.

As part of the initial setup, the Ministry will configure the licensees' subscription to APIs, and provide access to the APIC Developer Portal. The Developer Portal enables OBR support to complete the following tasks:

- generate and retrieve the value of the Client ID to be used by the licensee's consuming application
- identify API endpoints and the schema definitions for all APIs contained within the Product
- view analytics describing the types and volumes of API calls that the licensee has completed

Once connectivity/access are fully configured and functional, licensees will interact with the gateway directly by means of API calls.

All the required information including endpoints, API specifications, and connection information, will be provided by the OBR support team.

3.2 HTTP message headers

Message headers must be included in the REST calls to support both security and functionality. In addition to the standard HTTP headers needed to make REST requests, the OBR system expects an additional mandatory headers, the "Client ID", and the "Registertype".

3.2.1 Registertype header

Functional areas within the OBR application have been segmented to improve stability and performance of the business registry. As a result of this segmentation, REST messages must be routed to the appropriate application segment. To support this routing, the Registertype parameter must be added to the header of the HTTP request when calling a **subset** of the APIs. (Note that some APIs do not require the Registertype header.)

Example:

```
POST HTTPS://url-to-OBR.ws: HTTP/1.1
Accept-Encoding: gzip,deflate,br
Content-Type: application/json
... Other header values ...
Registertype: ...register-type...
... Other header values ...
```

3.2.2 Client ID header

The Client ID message header (‘*X-IBM-Client-Id*’) is mandatory and must be included in every API call. It acts as a security token to identify valid consumers of the APIs to the gateway platform. Any calls that do not contain a Client ID (or that use a Client ID that is invalid for a particular API consumer) will be rejected by the API gateway.

For more information regarding the Client ID header, refer to Section 5 ‘Security’.

Example:

```
POST HTTPS://url-to-OBR.ws: HTTP/1.1
Accept-Encoding: gzip,deflate,br
Content-Type: application/json
... Other header values ...
X-IBM-Client-Id: # Licensee's Client Id #
... Other header values ...
```

3.3 JSON message body

OBR APIs use REST-compliant JSON messages. The example below shows a sample REST request and resulting response.

REST Request Example:

```
{
   "OnbisEntitySearch" : {
      "SearchFor" : "1000014502",
      "ExactMatch" : "false"
   }
```

REST Response Example:

```
{
   "OnbisEntitySearchResponse" : {
      "status" : "Success",
      "ResultsPageSearchRow" : {
         ... Results overview ...
      },
      "EntityInfo" : [
         {
            ... Entity Information ...
         },
         {
            ... Entity Information ...
         }
      ]
   }
}
```

3.3.1 OpenAPI validation of incoming request messages

The OBR provides an OpenAPI 3 Schema Definition to API consumers that describes the required content and format of each REST message. Consumers should pre-validate their REST messages against the API definition before calling the APIs.

The API definitions will be provided to the subscribers.

3.4 Products, Plans and rate/burst limits

Products and Plans are proprietary packaging constructs that are used by the IBM API Connect tool. OBR uses Products to offer one or more API to application developers (API consumers) who will submit requests. OBR uses Plans to control access to the APIs and to manage API usage. Products are packages that contain both the APIs and the accompanying Plans, and they are used to manage the lifecycle of the APIs they contain.

Plans perform the following functions:

- Control which APIs a consumer can use
- Make available a collection of operations from one or more APIs
- Apply rate limits to APIs to differentiate between offerings

- Implement different rate limits to specify how many requests a consuming application is allowed to make during a specified time interval

4 Error management

4.1 Overview

As with any complex system, errors may occur. This section describes how errors will be handled by the OBR system and provides recommendations on how errors should be handled by the licensee's software. User/client software errors (e.g., UI-based field validation) should be detected and remediated by the client (licensee) software.

4.2 Categorization of error conditions

When calling OBR APIs, errors can arise from several factors such as communication failures, authentication failure, OBR system unavailability, and/or business-rule validation failures. Error conditions are discussed below in three categories:

- API faults
- System exceptions
- Business-rule exceptions

4.2.1 API faults

Communication errors can be returned by the API gateway for a variety of reasons, including:

- expired security certificate
- authentication issues
- API caller IP address not being included in the allow-access list
- malformed JSON In API requests
- invalid URLs/URIs
- exceeded limits on call volumes, etc. will be rejected by the gateway

When the API gateway rejects messages under these types of conditions, an API fault response is returned. These responses will include the appropriate HTTP status codes. A sample of possible responses that API consumer applications can expect to see are shown below.

HTTP Response Code	Possible Reason
400	Client attempted to call OBR service with malformed request, i.e., malformed header or JSON body
401	Client has not properly identified themselves, e.g., missing Client ID header
404	Requested URL/service name is not recognized by OBR gateway

422	The JSON message body does not pass API validation
429	Client has exceeded rate/burst limit of allowable call volumes.

Example of an API fault response:

```
{
   "httpCode" : "400",
   "httpMessage" : "Bad Request",
   "moreInformation" : "Invalid object syntax"
}
```

4.2.2 System exceptions

The business rules underpinning the Ontario Business Registry will evolve over time and the OBR is continually being enhanced through the addition of new features and functions, changes to existing functionality, and resolution of defects. Updates to the OBR system will require scheduled periods of system downtime so that releases can be deployed to both the testing and production environments.

Planned outages in the testing and production environments will be communicated to licensees in advance, with the expectation that licensee systems will not connect to the impacted environment during the outage duration. (For planned releases to the live OBR system in production, API access will be disabled during the release period.) Any API calls attempted during system downtime will result in the consumers receiving an HTTP status code of either 404 or 500.

It is important to note that licensees would experience similar behaviour if the OBR system were to experience an unscheduled outage. During recovery from an unplanned outage, access to the OBR (including APIs) may be restricted until the system is stabilized.

4.2.3 Business-rule exceptions

There may be situations where the OBR system will reject a request or return a failure code in the response payload even though API validation has succeeded. These instances are usually caused by issues with the data in the request JSON. For example, if submitted data does not match formatting rules or if the data is subject to constraints defined by the OBR business rules, the system will reject the transaction and inform the requestor of which elements triggered the error, along with a replay of the request payload.

For a more in-depth discussion of the business rules that apply to different OBR transaction types, refer to the following link for information on Notice of filing requirements: https://www.ontario.ca/page/ontario-business-registry-all-services#section-6.

5 Security

5.1 Overview

The Ontario Business Registry system uses several techniques to manage security for its APIs and to ensure that only valid requests from known parties can be submitted to the Registry. These techniques include use of X.509 certificates, preconfigured security headers in REST messages, and IP filtering.

5.2 Authentication

All OBR APIs are secured and no unauthenticated calls to the APIs are permitted.

Secure access to the APIs is enabled by means of standards-based public key certificates, specifically X.509.

Any licensee approved to integrate with OBR APIs will be provided with two X.509 certificates (one for production access and one for non-production access).

The Ministry will use the certificate to generate an internal security token that will identify and authenticate the API caller so that when a licensee makes any API request that includes the certificate, the transaction will be associated with the correct OBR Organization User account.

APIC will also automatically generate an identifier called the Client ID ('*X-IBM-Client-Id*') that uniquely identifies the licensee. Both the certificate and the Client ID string must be present in **every** API call from the licensee.

Requests will be rejected by the API gateway if:

- the request does not contain the correct certificate information
  and/or
- the request does not contain the licensee's correct Client ID as a header attribute.

5.3 IP filtering

In addition to authentication through the use of certificates, the Ministry limits access to the OBR APIs through the use of an 'allow-access' list.

Approved licensees will be required to submit to the Ministry in advance all IP address(es) from which calls to the OBR APIs will originate. These IP addresses will be inserted by Ministry technical staff into the Ministry's allow-access list so that the licensee's API calls can reach the API gateway.

Note that for production access, the Ministry's preferred configuration is for all API calls from each licensee to originate from a single IP address. If this constraint will be the source of significant integration challenges for licensees, the Ministry is willing to consult with each licensee to arrive at alternative configuration options that are satisfactory to both parties. For non-production access (i.e., UAT environment), licensees will be permitted to have multiple IP addresses from which API calls may originate.

6 Testing

6.1 Overview

For approved licensees, integration testing is performed in the Ministry's UAT environment. Test transactions must not be submitted in the live registry system.

Licensees should confirm the following:

- connectivity from the licensee's own network/application to OBR
- submission of transactions (API requests)
- ability to receive expected API responses from OBR
- successful execution in UAT of all business transactions that they intend to execute in the live system once they connect to the production APIs

For test cases that have specific data requirements, the Ministry can assist licensees with identifying or creating specific entity data that will enable successful completion of the test case.

For issues encountered during testing that licensees are unable to resolve by themselves, please send an email to obr.inquiries@ontario.ca.

Sign-off will be required from each licensee indicating successful completion of the full testing cycle before production access will be granted.

6.2 Service endpoints

Licensees will be able to retrieve API endpoints and definition from the information package, provided by the OBR support team, as mentioned in section 3.1. These endpoints will differ between the testing environment and the production environment.

After a licensee's IP address has been added to the allow-list (as described in section 5.3), the licensee should be able to test connectivity between their environment and the Business Registry. Licensees can use an off-the-shelf API testing tool such as ReadyAPI, Postman, etc. to verify that they are able to connect to and communicate with OBR endpoints.

6.3 End-to-end integration testing

This phase of testing involves licensees invoking the OBR APIs (with support from the Ministry, as required) to confirm that API requests are being transmitted successfully and correct results are being returned as required by the use case.

For true end-to-end testing, the Ministry recommends that the licensee configure their own application to submit the API requests with the required headers and JSON payloads in the Ministry's testing environment. The licensee should:

- validate message payloads against the OBR's API prior to making requests to the OBR system since any payload not adhering to the API specification will be rejected (as described in section 3.3.1);
- verify that the results received in an API response confirm the completion of a successful transaction; and
- confirm behaviour of their own application software when an error or unexpected situation occurs.

End of Document